Def^n: $G(s) \in \mathbb{R}^{p\times m}(s)$ ← real rational matrix

$\hat{G}(s) \in \mathbb{R}^{p\times m}(s)$ is called inverse of $G(s)$ if $G(s)\hat{G}(s) = \hat{G}(s)G(s) = I$.

$$G^{-1} \sim \begin{bmatrix} A - BD^{-1}C & -BD^{-1} \\ D^{-1}C & D^{-1} \end{bmatrix}$$

$\boxed{L12}$ 30/09/2020

Assume $G(s) \in \mathbb{R}^{p\times m}(s)$ is proper. Then we call a state-space model (A,B,C,D) such that $G(s) = \begin{bmatrix} A & B \\ C & D \end{bmatrix}$ a realization of $G(s)$.

e.g. $\frac{2}{s+1}$ ✓ $\xleftrightarrow{\text{realization}}$ $\dot{x} = -x + u$, $y = 2x$ $\begin{bmatrix} -1 & 1 \\ 2 & 0 \end{bmatrix}$ ✓, $\begin{bmatrix} -1 & 2 \\ 1 & 0 \end{bmatrix}$ ✓

e.g. A realization (A,B,C,D) of $G(s)$ is called minimal realization of $G(s)$ if A has the smallest dimension possible.

e.g. $G(s) = \begin{bmatrix} \frac{1}{s+1} & \frac{2}{s+1} \end{bmatrix} \in \mathbb{R}^{1\times 2}(s)$

$\mathcal{C}_{AB} = [B_1 \;\; A_1B_1]$ $\mathcal{O}_{CA} = \begin{bmatrix} 1 & 2 \\ -1 & -2 \end{bmatrix}$ Unobservable

✗ $n=2$ ✓ $A_1 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix}$, $B_1 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, $C_1 = [1 \;\; 2]$, $D_1 = 0$ $\quad [1 \;\; 2]\begin{bmatrix} s+1 & 0 \\ 0 & s+1 \end{bmatrix}^{-1}\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = G(s)$

$n=1$ ✓ $A_2 = -1$, $B_2 = 1$, $C_2 = [1 \;\; 2]$, $D = 0$ $\quad [1 \;\; 2]\frac{1}{s+1} = G(s)$

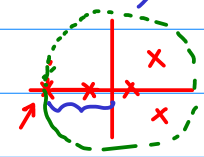
A state space realization (A,B,C,D) of $G(s)$ is minimal $\Longleftrightarrow$ (A,B) is controllable & (C,A) is observable

L16 08/10/2020

Lemma: Let A & B be two matrices of suitable dimension. Then

① $\rho(A) \leq \|A\|$ (This is also true for Frob. norm & other induced norms)

$\sigma(A) := \{\lambda_1, \lambda_2 \cdots, \lambda_n\}$ $\quad \lambda_i \in \mathbb{C}$ $\quad \rho(A) := \max_i |\lambda_i|$



② $\|AB\| \leq \|A\| \|B\|$ (This is true for any induced norm)

③ $\|UAV\| = \|A\|$ and $\|UAV\|_F = \|A\|_F$ for any appropriate dimension unitary U & V

④ $\underbrace{\|AB\|_F}_{\text{Frob}} \leq \underbrace{\|A\|}_{\text{induced 2-norm}} \underbrace{\|B\|_F}_{\text{Frob}}$ and $\|AB\|_F \leq \|B\| \|A\|_F$

Singular Value Decomposition: (SVD)

Theorem: Let $A \in \mathbb{C}^{m \times n}$. There exists unitary matrices

$$U = [u_1 \ u_2 \ u_3 \ \cdots \ u_m] \in \mathbb{F}^{m \times m}$$

$$V = [v_1 \ v_2 \ v_3 \ \cdots \ v_n] \in \mathbb{F}^{n \times n}$$

such that $A = U \Sigma V^*$, $\quad \Sigma = \begin{bmatrix} \Sigma_1 & 0 \\ 0 & 0 \end{bmatrix}$, $\quad \Sigma_1 = \begin{bmatrix} \sigma_1 & & & \\ & \sigma_2 & & \\ & & \ddots & \\ & & & \sigma_r \end{bmatrix}$

Singular values of A

$T^{-1} A T = J$

$\sigma(A) = \sigma(J)$

similarity transform

$$\boxed{U^* A V = \Sigma}$$

Not eigenvalues

diagonal matrix

and $\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_r \geq 0$.

$$A = U \Sigma V^* \Rightarrow AV = U\Sigma \Rightarrow A[v_1 \ v_2 \cdots v_n] = [u_1 \ u_2 \cdots u_m] \begin{bmatrix} \sigma_1 & & & & & \\ & \sigma_2 & & & & \\ & & \ddots & & & \\ & & & \sigma_r & & \\ & & & & 0 & \\ & & & & & \ddots \ 0 \end{bmatrix}$$

1st right singular vector

$$Av_1 = \sigma_1 u_1$$

1st left singular vector

1st singular value of A

$$Av_i = \sigma_i u_i \longrightarrow (1)$$

$$A^* u_i = \sigma_i v_i \longrightarrow (2)$$

$$(1) \Rightarrow A^* A v_i = \sigma_i A^* u_i = \sigma_i^2 v_i$$

$$A = U \Sigma V^*$$

$$\Rightarrow A^* = V \Sigma U^*$$

$$\Rightarrow A^* U = V \Sigma$$

$$\therefore * \quad \boxed{A^* A v_i = \sigma_i^2 v_i}$$

$$A A^* u_i = \sigma_i^2 u_i$$

σ_i^2 are eigenvalues of A^*A and v_i are corresponding rt. eig vectors.

$$\|A\|_2 = \sqrt{\lambda_{max}(A^*A)}$$

$$= \sqrt{\sigma_1^2}$$

$$\sigma(A^*A) = \{\sigma_1^2, \sigma_2^2, \ldots \sigma_r^2, 0 \ldots 0\}$$

$$\boxed{\|A\|_2 = \sigma_1}$$

Lemma: Let $A \in \mathbb{C}^{m \times n}$ and $\sigma_1 \geq \sigma_2 \geq \ldots \geq \sigma_r > 0$ $\quad r \leq \min\{m, n\}$

Then

1. $\text{rank}(A) = r$
2. $\ker(A) = \langle v_{r+1} \ldots v_n \rangle$
   $\ker(A)^\perp = \langle v_1, \ldots v_r \rangle$
3. $\text{img}(A) = \langle u_1, \ldots u_r \rangle$
   $\text{img}(A)^\perp = \langle u_{r+1}, \ldots u_m \rangle$
4. $A = \sum_{i=1}^{r} \sigma_i u_i v_i^*$ (Dyadic expansion)
5. $\|A\|_F^2 = \sigma_1^2 + \sigma_2^2 + \ldots \sigma_r^2$
6. $\|A\| = \sigma_1$

$$AV = U\Sigma$$

$$A[v_1 \ldots v_r \; v_{r+1} \ldots v_n]$$

$$= [u_1 \ldots u_r \; u_{r+1} \ldots u_m] \begin{bmatrix} \sigma_1 & & & & \\ & \sigma_2 & & & \\ & & \ddots & & \\ & & & \sigma_r & \\ & & & & 0 \\ & & & & & \ddots \\ & & & & & & 0 \end{bmatrix}$$

$$A = [u_1 \ldots u_r] \begin{bmatrix} \sigma_1 & & \\ & \ddots & \\ & & \sigma_r \end{bmatrix} \begin{bmatrix} - v_1^* - \\ - v_2^* - \\ \vdots \\ - v_r^* - \end{bmatrix}$$

$$= \sigma_1 u_1 v_1^* + \sigma_2 u_2 v_2^* + \ldots \sigma_r u_r v_r^*$$